Candidate's Name: ……

Signature: ……

Random No.	Personal No.

P530/2

BIOLOGY

PRACTICAL

Paper 2

2026

3 hours 15 minutes

INVESTIGATIVE BIOLOGY PRE-UNEB SET 3

Uganda Advanced Certificate of Education

BIOLOGY PRACTICAL

INSTRUCTIONS TO CANDIDATES

- *This examination paper consists of two items. Respond to both items.*
- *Write all responses in the spaces provided.*
- *Use sharp pencils for drawings. Coloured pencils should not be used.*
- *Show all your working for any calculation.*
- *No extra paper is allowed. Any work written on additional sheets will not be scored*.

Item No.	Code	Weighted Score	Scorers' Initials
1			
2			
Total			

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Turn Over

Item 1

Murchison Falls National Park manages a wildlife sanctuary. Local monitors in the park studied two small terrestrial vertebrates living among a dominant climbing plant that thrives even in the dry season and serves as a food source from its bark and habitat for the organisms. Numbers of one vertebrate declined sharply while the other remained stable. The community predicts that the difference in survival is linked to key processes in blood circulation, local sensitivity, predation risk, or the ability to locate surviving plants for food and shelter. They are investigating the cause of the difference in survival of the two animals by comparing their structures. Data were recorded for P, which is threatened and killed by flesh; Q has been provided with a stem piece of a surviving plant labelled R.

Table 1 Structural features of specimen P and specimen Q

Feature	**Specimen P**	**Specimen Q**
Number of heart chambers	3	
Number of main blood vessels taking and bringing blood to the heart	2	
Shape of earlobes	Flattened with thick fur	
Ratio of hind-limb length to fore-limb length	1:1	
Number of main joints in each limb	1	

Task:

(a) Obtain data from specimen Q and complete Table 1.

(Include a procedure for the dissection that displays the heart and the limbs.)

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(b) Make a drawing to show the relevant vascular structures displayed from your dissection.

(b) Prepare a transverse section of stem R, treat it with a suitable reagent, and examine it under the microscope. Make a drawing and label the tissues that conduct substances in the plant.

(c) Analyse the findings to account for the difference in survival between specimen P and specimen Q in the sanctuary, including the relevance of the tissue observed in stem R to the animals' nutrition.

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Item 2

Two neighbouring farmers in Pallisa District grow two leafy vegetables in separate gardens. One garden is watered from a borehole; the other receives water from a nearby drainage channel.

Both plants grow well in the borehole garden. In the drainage-water garden, one plant stays healthy while the other quickly wilts and dies.

Extension officers suggested the difference in survival might be due to differences in stomatal density on the upper leaf surface and the effect of dissolved solutes in the soil water on guard cell turgor towards stomatal opening. They collected leaves from both plants and samples of the water used in each garden.

You are provided with: a leaf **K** from a C_3 plant that dies in the drainage water garden, Leaf **L** from a CAM plant that survives in both gardens, Solutions **A** and **B** from both gardens, a Microscope

Task

(a) Design and carry out an investigation on leaves A and B to advise the farmers on how to maximise the productivity of the leafy vegetables.

Response

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CONFIDENTIAL

Great care should be taken that the information given below does not reach the candidates either directly or indirectly.

Each candidate will require:

- *One freshly killed rat, labelled Q*
- *Young dicot stem of (sunflower, pumpkin or Bidens), labelled R*
- *Leaf of Bryophyllum, labelled K*
- *Leaf of Commelina, labelled L*
- *30 cm^3 of 0.6 mol dm^{-3} sodium chloride solution or 0.8M sucrose, labelled Solution A*
- *30 cm^3 of distilled water, labelled Solution B*
- *Phloroglucinol and concentrated hydrochloric acid*
- *Dissecting kit, ruler, light microscope, slides and coverslips, 6 test tubes or small beakers, 10 cm^3 measuring cylinder, blotting paper, stop clock, knife or cork borer*

P530/2 BIOLOGY PRACTICAL — SCORING GUIDE

PRE-UNEB SET 3 | ITEM 1 = 12 ITEM 2 = 17

ITEM 1

Code DS — Dissection procedures

Expected response

- ✓ *Place specimen Q ventral side uppermost and pin through the feet to display the limbs.*
- ✓ *Make a mid-ventral skin incision, loosen and pin the skin back.*
- ✓ *Cut the abdominal wall on the midline from the diaphragm to the pelvic girdle and pin the wall back*
- ✓ *Cut out the alimentary canal by ligaturing the coeliac artery, hepatic portal vein, anterior mesenteric artery, and cut them distal to ligatured points. Remove the alimentary canal*
- ✓ *Cut the rib cage laterally on both sides towards the neck and remove the ventral view of the rib cage.*
- ✓ *Cut out the thymus gland to display major anterior blood vessels from the heart and its anterior part like anterior vena cava, common carotid artery.*
- ✓ *Cut away the pericardium to clearly display the heart and major vessels.*
- ✓ *Displace the left lung anteriorly to the right to display the dorsal aorta and posterior and anterior vena cava branching and pulmonary vessels.*

Bases of assessment

Level	What is seen on the script	Weighted score
Ap	Four or more display or displacement steps that expose the heart, vessels leaving it, and allow examination of limbs and earlobes.	3
Ae	Two or three display steps, including opening of the thorax.	2
Al	Only orientation, pinning, skinning or opening the body wall without displaying the heart.	1
0	Thorax not opened or no procedure.	0

Code DM — Measurements

Expected response

Feature	Specimen P	Specimen Q

Number of heart chambers	3	4
Number of main blood vessels taking away from and bringing blood to the heart	2	Accept 6 to 8
Shape of earlobes	Flattened with thick fur	Funnel-shaped with thin fur
Ratio of hind-limb length to fore-limb length	1:1	2:1
Number of main joints in each limb	1	2

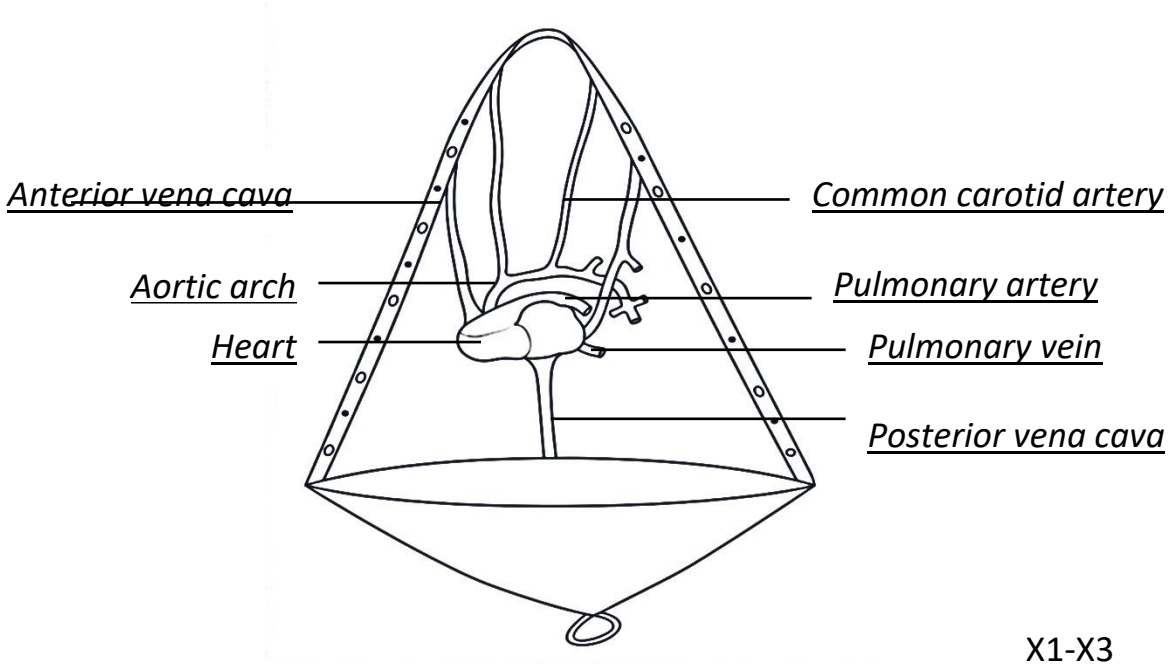
Bases of assessment

Level	***What is seen on the script***	***Weighted score***
Ap	*Three or more of the five entries are correct and comparative with P.*	*3*
Ae	*Exactly two entries are correct and comparative.*	*2*
Al	*Exactly one entry correct and comparative.*	*1*
0	*No correct comparative entry.*	*0*

Code DR — Drawing

Expected response

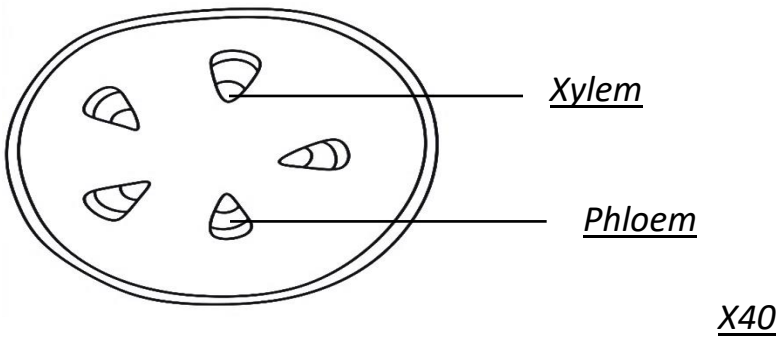
(b) *Drawing showing main blood vessels taking away from and bringing back blood to the heart of specimen Q.*



X1-X3

(c)

Drawing showing a transverse section through specimen R to show vascular tissues



X40

Bases of assessment

Level	***What is seen on the script***	***Weighted score***
Ap	*6 or more of the 8 attributes present.*	*3*
Ae	*4 or 5 attributes.*	*2*
Al	*1 to 3 attributes.*	*1*
0	*No usable drawing or coloured pencil used.*	*0*

Code ST — Suitability

Expected response

- ✓ *Q has four heart chambers, which generate higher pressure for quick delivery of blood to body muscles for efficient climbing and separate oxygenated blood from deoxygenated blood, unlike P.*
- ✓ *Q has 3 major vessels that drain the heart, and 3 major vessels supply blood to the heart for a more efficient delivery of nutrients and oxygen to the body muscles used in climbing to survive plants for food and shelter, unlike P.*
- ✓ *The hind limb of Q is two times longer than the forelimb (2:1 ratio) to provide larger propulsive force, while the forelimb is shorter to absorb shock during movement to swiftly reach and grip stems for their survival, unlike P.*
- ✓ *Earlobes of Q are funnel-shaped to efficiently collect sound waves to easily escape predators, unlike P with flattened lobes that do not collect waves.*
- ✓ *Little hairs on the earlobes reduce insulation against heat loss, thus losing more heat in a hot environment, while thick fur in P increases insulation.*

- ✓ *2 joints in the hind limb increase flexibility during climbing to feed on the bark, unlike a single joint, which limits flexibility.*
- ✓ *Phloem lies on the outer side of the vascular bundles in stem R and contains organic substances; Q can access this outer tissue for nutrition while P cannot*

Rubric

Level	***What is seen on the script***	***Weighted score***
Ap	*Five links, each quoting data or the drawn tissue and giving its significance for survival.*	*3*
Ae	*Three such links.*	*2*
Al	*One such link.*	*1*
0	*Story restated without using Table 1 or the drawing.*	*0*

ITEM 2

Code PE — Planning of the experiment

Expected response

Aim. *To determine the stomatal density of upper-surface of Leaves K and L, and the effect of Solutions A and B on stomatal opening and closure, to identify the causes of wilting and death of some plants when watered with drainage water, and provide recommendations to Pallisa farmers to maximize vegetable yields.*

Hypothesis. *Leaf K has a higher stomatal density on the upper surface than leaf L, leading to greater water loss, and solution A causes stomatal closure while solution B causes stomatal opening because guard cells lose water to concentrated solution A and gain it from dilute solution B by osmosis.*

Variables.

Independent: *Type of leaf (K or L); type of solution (A or B).*

Dependent: *Stomatal density; opening and closure of stomatal aperture*

Controlled: *Area counted/Field of View, size of tissue used, volume of solution used, time of immersion of epidermal peels in solutions A and B.*

Bases of assessment

Level	What is seen on the script	Weighted score
Ap	Aim names both comparisons and the purpose; hypothesis gives a prediction and reason for each factor; both pairs of variables stated. At least three accurate points.	4
Ae	No Ap. What is to be done without the purpose, or two predictions without reasons.	2
Al	Only one independent variable, or one prediction with one reason.	1
0	Not taken from the Pallisa farmers' story.	0

Code ME — Methods of the experiment

Expected response

Risks.

- *Microscope stains can irritate eyes in case it gets into contact — avoid contact by wearing protective gloves and avoid touching eyes before washing hands.*
- *Knife/razor blade used to cut the body if not carefully used to peel — cut and peel the tissues carefully.*

Procedure.

Determining stomatal density of K and L

- ✓ *2 Epidermal pieces of the upper surface of leaves K and L were peeled off and dipped in distilled water in a watch glass.*
- ✓ *Using forceps, each of the peeled pieces was transferred to a microscope slide in a drop of water.*
- ✓ *A cover slip was used to cover an epidermal layer on the microscope slide and transferred to the microscope stage for observations.*
- ✓ *The number of stomata was counted for each epidermal peel, and results were recorded in the same Field of view.*
- ✓ *Using a long ruler, the diameter of the field of view was measured and used to determine the area of the field of view using the formula* πr^2
- ✓ *Using the formula* $\frac{Number\ of\ stomata}{Area\ of\ field\ of\ view}$

Effect of solution A and B on stomatal aperture

- ✓ *Two Petri dishes were labelled A and B.*
- ✓ *$10cm^3$ of each of the solutions A and B were measured and added to the respective Petri dish.*

- ✓ *Two fresh epidermal peels were prepared from leaf L, and each was transferred into Petri dishes A and B.*
- ✓ *The stopwatch was immediately started, and the setups were left for 10minutes.*
- ✓ *After 10 minutes, stomatal aperture was examined under medium power of the microscope from each of the pieces of epidermal layer from A and B.*
- ✓ *Results were recorded.*

Bases of assessment

Relevance: Uses both leaves and both solutions.

Accuracy: Equal areas or equal tissue sizes, stated volumes and time.

Coherency: Prepare, count, observe stomatal aperture.

Level	What is seen on the script	Weighted score
Ap	Two or more Ap across relevance, accuracy, and coherency.	4
1 Ap	Exactly one Ap.	3
Ae	Risks with mitigations, or two accurate or two coherent points.	2
Al	One risk with mitigation, or one accurate point, or one coherent link.	1

Code DA — Data presentation

Expected response

Table showing stomatal density of leaves K and L

Leaf	*Number of stomata*	*Area of field of view using πr^2*	*Stomatal density (stomata/mm)*
K	*27*	*4*	*7*
L	*26*	*4*	*110*

Table showing the effect of solutions A and B on the stomatal aperture

Solution s	*Appearance of stomatal aperture*
A	*Closed aperture*
B	*Opened aperture*

Rubric

Level	*What is seen on the script*	*Weighted score*
Ap	*Both the stomatal comparison and the aperture comparison are presented and show the expected direction.*	*3*
Ae	*One comparison fully presented.*	*2*
Al	*Results given only in sentences.*	*1*
0	*No results.*	*0*

Code AN — Explanation of results

Expected response

Variation in stomatal density

Leaf L has a significantly higher stomatal density on its upper surface compared to Leaf K. This provides a larger overall surface area for water loss through transpiration, making Leaf L highly vulnerable to rapid dehydration when environmental water uptake is low, while K has a lower stomatal density, thus lower transpiration.

Effect of solutions A and B

Solution A (Drainage Water) *contains higher dissolved solute levels (hypertonic relative to cell sap) than the cell sap of epidermal cells. Water moves out of the guard cells into Solution A by* ***osmosis****. Guard cells lose turgor, become flaccid, and collapse inward, closing the stomatal aperture. While closure reduces transpiration, it prevents* CO_2 *uptake, leading to wilting, disrupting photosynthesis, and eventual death in Leaf K and the whole plant.*

Solution B (Borehole Water) has *a lower solute concentration (hypotonic) than cell sap. Water enters guard cells by* ***osmosis****, making them turgid and stretching outward to open the stomatal pore for more carbon dioxide uptake. This allows active photosynthesis and healthy growth for both plants.*

Bases of assessment

Level	What is seen on the script	Weighted score
Ap	Both the stomatal-density effect and the osmotic effect of A versus B are explained and linked to the results and the farmers' observations. If links are below 3, Ae is the maximum.	3
Ae	One factor fully explained and linked.	2
Al	Stomata or osmosis named but not linked to the data.	1
0	No explanation.	0

Code CR — Conclusion and recommendation

Expected response

Conclusion. *Leaf L has a higher upper-surface stomatal density than Leaf K, and exposure to high-solute Solution A causes guard cells to lose turgor and close stomatal pores, whereas dilute Solution B maintains high guard cell turgor and wide stomatal apertures.*

Recommendation.

- *Plant the resilient crop with (**Leaf K**) in the garden supplied by drainage-channel water, as its low upper stomatal density and guard cell adaptation due to close stomata without disrupting photosynthesis during day (CAM plant) withstand osmotic stress, unlike the plant from which leaf L was obtained.*
- *Utilize **borehole water (Solution B)** for both crops whenever possible instead of A from the drainage channel, as its lower salt content maintains guard cell turgidity, keeps stomata open for gas exchange, and maximizes overall vegetable productivity.*

Bases of assessment

Level	What is seen on the script	Weighted score
Ap	Both independent variables appear correctly in the conclusion; the recommendation names leaf L for the saline garden and solution B as the preferred water.	3
Ae	One independent variable and the dependent variable are correct in both the conclusion and the recommendation.	2
Al	One independent variable is correct in either the conclusion or the recommendation.	1
0	Contradicts the candidate's results, or absent.	0

END OF SCORING GUIDE